FURPS+ Requirements for RP-Cloud Labs

The following six requirements are for the Tool-Session Service of RP-Cloud Labs, which allows students to access hosted tools through a browser without installing them on their computers.

No.	Category	Requirement
1	Functionality	The system shall allow a student with a validated account to request a tool session only when their college has available software licence capacity.
2	Usability	The system shall provide a simple browser-based interface that allows students to request a tool session and view its status without installing additional software.
3	Reliability	The system shall prevent the number of active sessions for a college from exceeding its available licence capacity, including when multiple students request sessions simultaneously.
4	Performance	The system shall return a response to a valid session request within two seconds under specified normal operating conditions, excluding the time required to stream the hosted tool itself.
5	Supportability	The system shall record session requests, failures, and errors in logs to help administrators diagnose problems and maintain the service.
6	Scalability (additional non-functional requirement)	The system shall support increasing numbers of concurrent tool sessions by allowing additional service instances to be deployed without requiring changes to the existing application code.

Why these requirements are important

- Functionality: Ensures students can request the tools they need.
- Usability: Makes the service easy to use, even on low-spec computers.
- Reliability: Prevents licence limits from being exceeded.
- Performance: Ensures requests receive responses quickly.
- Supportability: Helps administrators identify and fix errors.
- Scalability: Allows the service to handle more students and concurrent sessions as additional colleges join RP-Cloud Labs.